
Persistent Data-Order Effects Precede Directional Alignment in Language Models

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Abstract

Training a language model on the same data in a different order produces a different model. We ask when this difference appears in the model’s predictions, and when the difference between the two runs’ weights settles into its final direction. We use public Pythia-160M checkpoints from runs that share an initialization and differ only in data order. At each checkpoint we compare the divergence between the two runs’ next-token predictions with the cosine between their current and final weight differences. At about one-ninth of training, the order-only prediction gap is already 93% of the gap between two runs that also differ in initialization. The weight difference at that point is still far from its final direction, with a cosine of 0.29. Under the same learning-rate schedule, single runs and the joint pair also align slowly. Most of this lag therefore reflects training in general and not data order. We then test whether the order effect can be measured as recoverable, using block-wise weight grafts and Fisher-based local metrics. Both fail validity checks fixed in advance, so we make no recoverability claim. All comparisons use one model size, three correlated order-only pairs, and one evaluation shard from the training distribution.

1 Introduction

Training applies an ordered sequence of updates. Two runs that start from the same weights and see the same data can still end at different models. Order sensitivity itself is established [5, 6]. Less is known about how the effect develops during pretraining. Does an order-induced difference in predictions persist to the end of training? When does it reach its final size, and when does the weight difference that carries it stop changing direction? Can the effect be corrected, and can a proposed recovery measurement be trusted?

A scalar output divergence need not track the direction of the weight difference. Later updates can rotate a displacement while its effect on predictions stays the same size. We study this using public Pythia and PolyPythias trajectories [1, 2], without new pretraining. The data-seed variants hold initialization fixed and change only data order, which few pretraining-scale studies can do.

Our contributions are the following.

1. **Prediction gaps reach full size before weight directions settle.** At step 16,000 of 143,000, the order-only prediction gap is 93% of the joint initialization-and-order gap. The raw weight difference at that step has a cosine of only 0.294 with its final direction. A schedule

control shows that most of this directional lag also appears in single runs and in the joint pair (Section 3).

2. **Order-only prediction differences persist.** Three correlated order-only comparisons end at 0.290 to 0.293 nats of symmetric KL, close to the joint pair’s 0.306.
3. **Recovery measurements fail validity checks.** The tested graft class and Fisher estimators fail criteria frozen before the runs. We report these as reasons to withhold recovery claims. They are not evidence of irreversibility.

2 Framework and measurement design

Order defects. This section fixes vocabulary. None of its quantities is measured directly in this paper. Let x include parameters and any optimizer state, and let U_i be one update map. The adjacent-swap defect is

$$\delta_{ij}(x) = U_j(U_i(x)) - U_i(U_j(x)). \quad (1)$$

For smooth SGD updates $U_i(x) = x - \eta g_i(x)$, $\delta_{ij}(x) = \eta^2(Dg_j(x)g_i(x) - Dg_i(x)g_j(x)) + O(\eta^3)$, which is the commutator term studied in prior work [5, 6]. Adam requires the augmented state, so this parameter-only expansion does not apply to it unchanged.

Connect two update orders by adjacent swaps. For swap k , let z_k be the state after its first ordering, $z_k + \delta_k$ the state after the alternative, and A_k the remaining shared updates. Then

$$\Delta_k = A_k(z_k + \delta_k) - A_k(z_k), \quad x_T^{(B)} - x_T^{(A)} = \sum_k \Delta_k, \quad (2)$$

by telescoping over intermediate orders. To first order $\Delta_k \approx DA_k(z_k)\delta_k$, so later dynamics can amplify, shrink, or rotate each local defect. This is why endpoint size and endpoint direction can evolve on different timescales. The checkpoint differences below are observed directly and are not decomposed into per-swap terms.

A conditional correction floor. For a parameter gap Δ , a fixed linear correction subspace S , and a fixed quadratic metric $\Sigma \succeq 0$, define

$$r_S(\Delta) = \min_{c \in S} \|\Sigma^{1/2}(\Delta - c)\|_2. \quad (3)$$

Every correction in S leaves a residual of at least $r_S(\Delta)$. The minimum exists because $\Sigma^{1/2}S$ is a finite-dimensional subspace. This floor is conditional on S and Σ . It is not a nonlinear irreversibility theorem or a computational lower bound. Section 4 tests whether a usable Σ exists, and no floor is evaluated.

Models, metric, and controls. Write m, d_1, d_2 for the main Pythia-160M run and its two data-seed variants, which share an initialization. We analyze $P_1 = (d_1, d_2)$, $P_2 = (m, d_1)$, and $P_3 = (m, d_2)$. These are three correlated pairs from three runs, and each run appears in two pairs. The joint pair J uses the two full-seed variants, which change initialization and order together. The endpoint is $T = 143,000$ updates, or 300B training tokens.

For next-token distributions on a frozen shard E , the output readout is

$$D_E(a, b) = \frac{1}{2|E|} \sum_{x \in E} [\text{KL}(p_a(\cdot|x) \| p_b(\cdot|x)) + \text{KL}(p_b(\cdot|x) \| p_a(\cdot|x))]. \quad (4)$$

The shard has 48 non-overlapping 2,048-token blocks from NeelNanda/pile-10k, joined with EOS separators under the Pythia tokenizer, for 98,304 prediction positions. It samples the training distribution and is not a held-out benchmark. Forward passes run in float32 with float64 accumulation. We use 12 checkpoints from step 0 to T and exclude step 1 after a checkpoint-integrity audit. A candidate initialization-only pair failed continuity checks and was excluded before analysis, so $D_E(J) - D_E(P_i)$ mixes the initialization effect with its interaction with order.

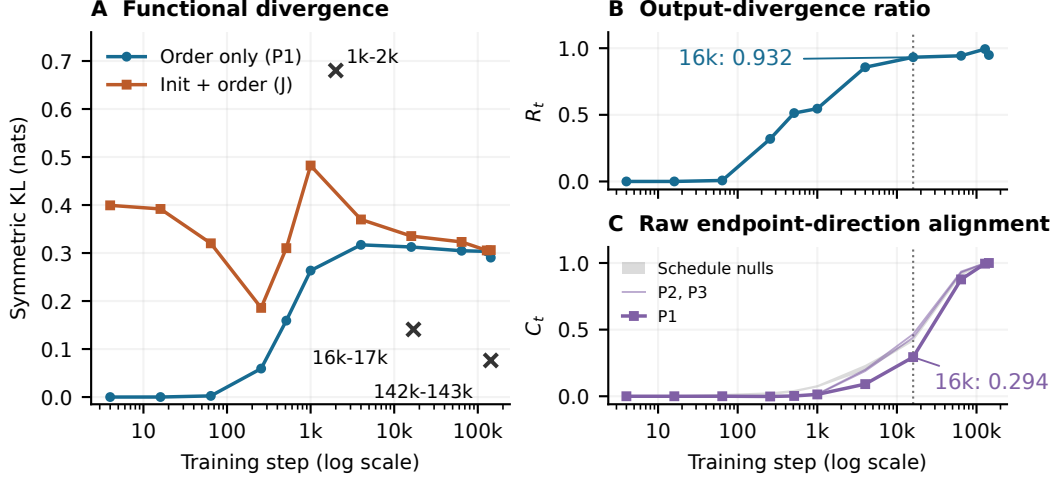


Figure 1: **Output divergence and raw weight-direction alignment.** (A) Symmetric KL for P_1 and the joint pair. Crosses mark same-run drift over the labeled windows and are not error bars. (B) Ratio $R_t = D_E(P_1, t) / D_E(J, t)$. (C) Raw cosine C_t between the weight difference at step t and at T . The grey band spans five schedule controls (four single-run displacements and the joint pair with its initial difference removed). Thin lines are P_2 and P_3 . Lines connect recorded checkpoints and are not fits. Ratio and cosine measure different objects, so their values are not directly comparable.

Table 1: Final-checkpoint divergence on the training-distribution shard. The three order-only pairs share one initialization and each run occurs twice. All ratios share the single joint-pair denominator.

Pair	D_E (nats)	$D_E / D_E(J)$	$D_E / \text{drift reference}$
P_1 : data-seed1 / data-seed2	0.290461	0.9489	3.81
P_2 : main / data-seed1	0.290638	0.9494	3.81
P_3 : main / data-seed2	0.293055	0.9573	3.84
J : full-seed1 / full-seed2	0.306112	1.0000	4.01

3 Prediction gaps reach full size before weight directions settle

Order-only prediction differences persist. The final order-only divergences are 0.290 to 0.293 nats, against 0.306112 for J (Table 1). All lie above the late same-run drift reference of 0.076275 and below the 160M versus 410M comparison of 0.460360. The largest is 1.009 times the smallest, under the predeclared threshold of 2.0. This agreement describes three correlated pairs and is not a confidence interval.

The drift reference is a descriptive comparison and not a noise floor. Over 1,000-update windows it is 0.680225 at steps 1,000 to 2,000 and 0.140831 at 16,000 to 17,000. Pair values exceed it only late in training, so separation from drift is a late-training statement.

Magnitude saturates before raw direction aligns. For P_1 , let $\Delta_t = \theta_{d_1, t} - \theta_{d_2, t}$ and $C_t = \langle \Delta_t, \Delta_T \rangle / (\|\Delta_t\|_2 \|\Delta_T\|_2)$. At step 16,000, $D_E(P_1) = 0.312597$ nats, already near its final value, and $R_t = 0.932$. The cosine at the same step is only 0.294 (Figure 1). This holds in absolute KL without the joint denominator. At 64,000, $R_t = 0.943$ and $C_t = 0.877$. The first sampled $C_t \geq 0.9$ occurs at 128,000. Crossing steps are bounds set by the checkpoint grid and are not stabilization times. P_2 and P_3 show the same pattern, with cosines of 0.441 and 0.466 at 16,000 and first crossings at 64,000.

Most of the directional lag comes from the schedule. A low cosine against the endpoint is partly expected for any trajectory. The endpoint gap includes every remaining update, and at step 16,000 the learning rate is still high. We therefore computed five schedule controls on the same grid. Four are single-run displacements, $\cos(\theta_t - \theta_0, \theta_T - \theta_0)$ for d_1 , d_2 , and both full-seed runs. The fifth is

Table 2: Local-metric certification on one endpoint pair. Each entry must be at most 0.25. The predictive-Fisher diagonal uses Monte Carlo sampling. A full empirical-Fisher diagnostic also fails all three probes.

Estimator	Random	Update	Order gap	Passed
Diagonal empirical Fisher	1.11	0.77	0.47	0/3
Per-tensor empirical Gram	1.90	0.61	1.31	0/3
Diagonal predictive Fisher	0.524	0.811	0.126	1/3

the joint pair with its initial weight difference subtracted. These control curves and the decision rule were registered before the computation.

At 16,000 the five controls lie between 0.411 and 0.432, and all first reach 0.9 at 64,000. Directional alignment is therefore slow for every trajectory under this schedule, so most of the low cosine reflects the schedule. P_1 lies 0.12 below the lowest control and clears our predeclared 0.10 margin. P_2 and P_3 lie inside or above the control band. Only one of three order-only pairs aligns more slowly than the controls, so we do not claim that data order delays alignment. The supported statement is narrower. Order-only prediction gaps reach nearly their full size while raw weight directions are still far from settled, and generic trajectories share this lag. Raw cosine depends on the parameterization and does not identify an output-space direction.

4 Recovery claims require validated probes

Parameter-type grafts leave the allowed loss region. At T we form hybrids $\theta_h = \theta_{d_1} + \alpha M_b(\theta_{d_2} - \theta_{d_1})$, where M_b selects a parameter-type block and $\alpha \in \{0.25, 0.5, 1\}$. A frozen rule rejects any hybrid whose token cross-entropy exceeds the worse parent’s by more than 0.5 nats, a threshold of 3.001560. Instrument checks passed, including reproduction of the parent divergence to relative error 1.4×10^{-10} .

Seven of eight complete block swaps are invalid. The loss band rejects 23 of 35 evaluated graft readouts, a count that includes control and second-anchor readouts of the same hybrids. Attention QKV carries 87% of the squared parameter gap and is invalid at every tested nonzero strength. Only internal LayerNorm stays valid through a full swap, and that hybrid is degraded and far from both parents. The frozen abort rule then stops the remaining controls, so spatial recoverability is inconclusive. This graft class fails at the tested strengths. We did not run the standard mode-connectivity control of interpolating all parameters at once [4]. These failures therefore cannot be attributed to a loss barrier between the endpoints.

Local Fisher metrics do not certify. To approach Eq. (3) without grafts, we test $D_E(\theta, \theta + v) \approx v^\top \Sigma_E v$ with $\Sigma_E = \frac{1}{2} \hat{F}$. Under our half-sum KL convention the predictive Fisher gives the local quadratic approximation. It is not in general the observed-label loss Hessian, and the empirical Fisher differs from both [3].

The frozen gate requires a relative prediction error of at most 0.25 on three scaled probe directions. These are a random direction, the update between the last two checkpoints, and the endpoint order gap. The probes raise cross-entropy by 0.0050, 0.0337, and 0.0087 nats. No estimator passes all three (Table 2), so the planned projection ladder and Fisher-weighted timing analysis were not run.

The predictive diagonal is within 13% along the endpoint gap but underpredicts the update-direction KL by about 5.3 times. *Agreement along one chosen direction does not certify a local metric.* Missing cross-parameter curvature fits this pattern, and diagonal estimators are expected to miss it. Independent builds reproduce the pass on the order gap and the failure on the random direction. The direct divergence readout reproduces to relative error 1.42×10^{-10} . We did not compute the exact predictive-Fisher quadratic form, which a Jacobian-vector product would give. Estimator error therefore remains confounded with a breakdown of the quadratic approximation at these step sizes. No tolerance was relaxed.

5 Discussion and limitations

Order-only prediction differences persist and reach nearly their full size while raw weight directions are still far from their endpoint. A schedule control shows that the slow directional alignment is mostly a property of training under this schedule. Output divergence, raw weight direction, local functional geometry, and actual correction are distinct quantities. None of them stands in for another without a validity check.

The evidence has specific boundaries. The three order-only pairs share one initialization, so they bound pair-to-pair noise but say nothing about variation across initializations. Independent runs would remove this limit. No initialization-only control exists for these checkpoints, so the joint comparison mixes initialization with its interaction with order. All divergences use one training-distribution shard. Held-out cross-entropy, calibration, or downstream tasks would show whether the endpoint differences matter for generalization. Raw cosine is sensitive to scale and permutation symmetries, and timing is limited by a sparse checkpoint grid. All results come from one model size and one training recipe.

Two experiments follow directly. Interpolating all parameters between the order-only endpoints would separate a loss barrier from the failure of block mixing. An exact predictive-Fisher quadratic form along each probe would separate estimator error from breakdown of the quadratic regime. A metric with cross-parameter structure should be certified on directions outside its represented span before any projection or recovery claim.

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